

साक्षी

Saakshi

A resilient, provable computer-based testing ecosystem

Hackathon Challenge 6 · Innovation & Differentiation

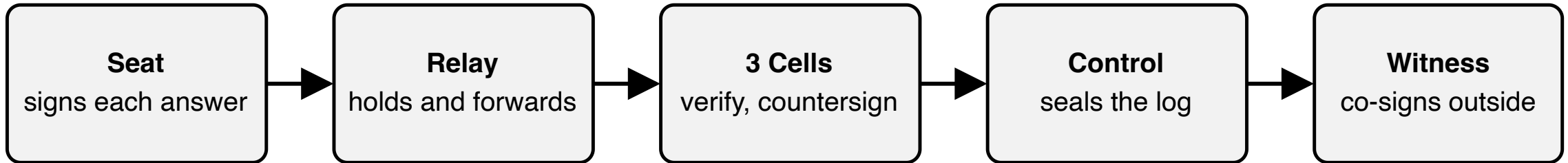
Team Kawaai: Anirudh, Bhumika, Arjav

A typical CBT today, and Saakshi

Most exam systems try not to fail. Saakshi assumes things will fail, and makes every failure safe, small and provable.

Question	Typical CBT today	Saakshi
Can a candidate prove what they answered?	No. Trust the server record.	Yes. A receipt code the candidate can check.
Can the database be edited quietly?	Yes. An insider edit may go unseen.	No. Every answer is signed and chained. An edit is located.
Who checks the log itself?	The same operator who keeps it.	An outside witness co-signs it.
Link goes down?	Answers can be lost or the exam stops.	Seat keeps going and sends later. 0 answers lost.
Centre cannot reach the server at the start?	Candidates wait. The centre may be cancelled.	A phoned offline code starts the exam.
After an outage?	Re-test everyone at the centre.	Re-test only the harmed candidates.
Suspicious pattern?	Auto penalty or a guess.	A flag goes to a human. History only backs it up.

One chain, checked at every step



0

answers lost in 16,000 (20 of 20 runs)

1,000

of 1,000 edits located

~1 s

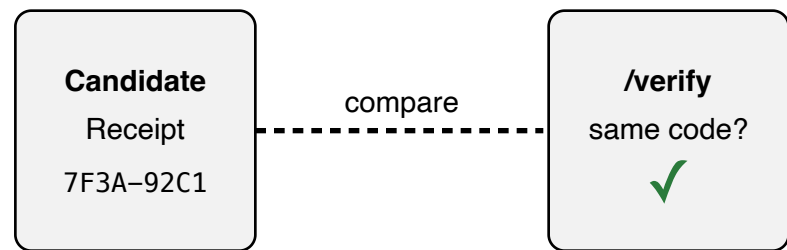
recovery after a cell loss

1,190

candidates Spared (synthetic)

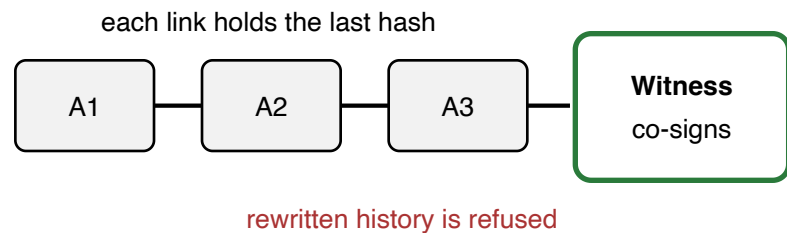
Honest limit: a seat signature proves a record was not changed after the device made it. It does not prove the candidate chose it. All analytics numbers here come from synthetic data.

Proof and tamper evidence



1. A receipt the candidate can check

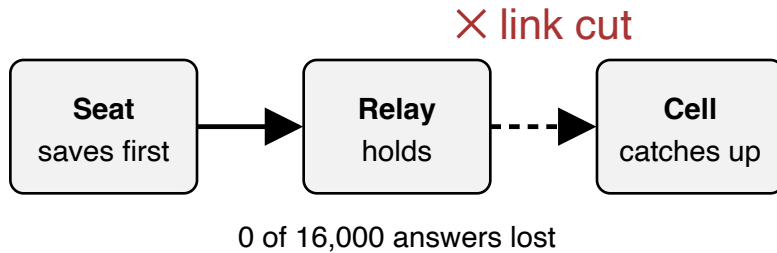
The seat makes the receipt code at submit, even offline. It equals the code the cell countersigns. The candidate and the public page can compare them.



2. A signed log with an outside witness

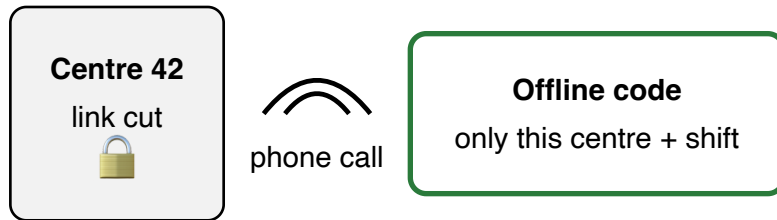
Answers are chained, and the log is sealed. A separate witness only co-signs a log that grows from the last one it saw. In our test, a rewritten 94-entry log was refused and raised a tamper alert. 1,000 of 1,000 byte flips were located.

Nothing lost, offline start



3. Answers never lost

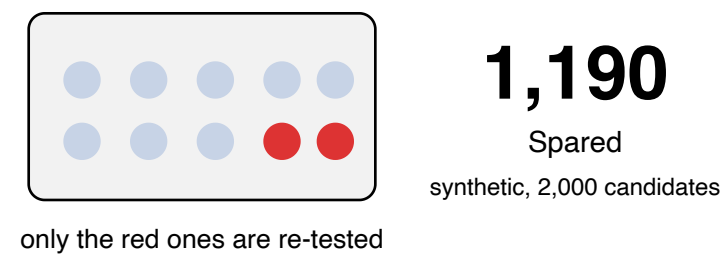
Every answer is saved on the seat first, then forwarded. If a link or a cell fails, the data waits and catches up. In 20 of 20 chaos runs, 16,000 sent equalled 16,000 stored. Recovery took about 1 second.



4. An offline code start

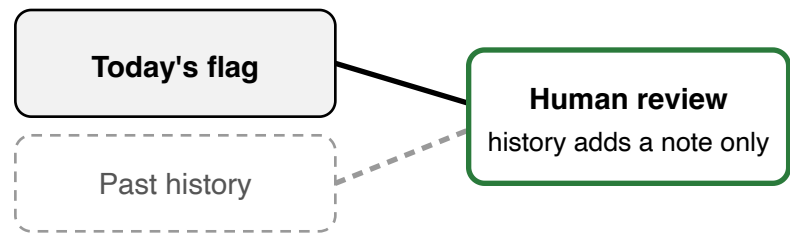
Two of three key holders release the paper keys. A centre with no link gets a code by phone. The code works only for that centre and shift. A typo, or another centre's code, is refused. Result: 308 candidates at 7 centres started.

Fair, human-led decisions



5. Re-test only the harmed

The cells' own records show who lost time. Only those candidates are re-tested. In the demo, 193 were re-tested and 1,190 were spared a needless re-test.



6. History only corroborates

Past records never create a flag and never change a decision. They only add a note. In the demo the flag set and the headline were unchanged, with 53 flags, 4 of them corroborated and none on an honest candidate.

How each idea changes the exam day

Moment	Old way	Saakshi way
Before	Hope the software is clean	Gate blocks remote tools by name (AnyDesk and an overlay tool were blocked)
Start	Offline centre is stuck	Phoned offline code unlocks it
During	Outage means lost work	Save first, send later
After	Disputes cannot be settled	Receipt and signed log settle them
Decide	Re-test a whole centre	Re-test only the harmed

No candidate is ever penalised by the system on its own. A person reviews every flag.

Tested on macOS and Windows. Code and evidence: <https://github.com/anirudh12032008/saakshi>